

Polkadot Playground

A library, a CLI, and a registry, so you can ship an app on the Polkadot stack and share it with anyone.

By the end of this session, you will have everything needed to ship to Polkadot.

01 Install one CLI.
`dot`

02 Set up your account.
`dot init`

03 Remix an existing app, or start fresh.
`dot mod · @polkadot-apps`

04 Deploy end-to-end. Optionally publish.
`dot deploy → playground.dot`

No servers, no Vercel. The bytes live on the Bulletin Chain, the name on DotNS, the ownership on Asset Hub.

Three components. One loop.

01 · LIBRARY

@polkadot-apps

14 TypeScript packages. Typed chain access, signing, Bulletin uploads, Statement Store, contracts.

02 · CLI

playground-cli

One binary: dot. Host login, account funding, account mapping, bulletin attestations, build, deploy, mod, auto-update.

03 · REGISTRY

playground-app

A registry browser at playground.dot. Discover, open, and remix other people's apps.

Library → CLI → registry. You write against the library, ship with the CLI, publish to the registry.

@polkadot-apps, the primitives for building apps.

[@polkadot-apps/chain-client](#) typed multi-chain client

[@polkadot-apps/signer](#) Host / extension / dev, one API

[@polkadot-apps/statement-store](#) pub/sub over statements

[@polkadot-apps/host](#) Desktop / Mobile detection

[@polkadot-apps/crypto](#) NaCl, KDF, symmetric

[@polkadot-apps/storage](#) k/v with host auto-detect

[@polkadot-apps/utils](#) encoding, token formatting

[@polkadot-apps/tx](#) submit + watch tx lifecycle

[@polkadot-apps/bulletin](#) upload + fetch on Bulletin

[@polkadot-apps/contracts](#) typed Solidity + ink! calls

[@polkadot-apps/address](#) SS58 ↔ H160 utilities

[@polkadot-apps/keys](#) HD derivation + session keys

[@polkadot-apps/descriptors](#) pre-built chain descriptors

[@polkadot-apps/logger](#) namespace-filtered logs

Framework-agnostic. Tree-shakeable. Consistent API for building for the Polkadot Triangle.

One binary. Five commands.

```
curl -fsSL https://raw.githubusercontent.com/paritytech/playground-cli/main/install.sh | bash
```

- dot init** login + toolchain + test funds + account mapping + bulletin authorisation · runs once
- dot build** detects pnpm / yarn / bun / npm, runs install and build
- dot deploy** Bulletin → DotNS → (optionally) playground registry
- dot mod** fork a published app into your GitHub
- dot update** self-update from GitHub releases

Initialise toolchain and login with Host account.

dot init · polkadot playground · paseo v0.7.0

✓ logged in 5GeC1iH9H418mLYdKMB9dCXrtckc7XErVacM1GagcLv6Jcik

dependencies

✓ rustup
✓ Rust nightly
✓ rust-src
✓ cdm & cargo-pvm-contract
✓ IPFS
✓ GitHub CLI
✓ authenticated

account

✓ funding 8.15 PAS
✓ mapping mapped
✓ bulletin >30d · #2,332,220
1000 txs · 100 MB remaining

✓ setup complete

01 Prints a QR code to your terminal.

02 Scan it with Polkadot Mobile App. Approve the session.

03 In parallel: rustup, nightly, rust-src, cdm, IPFS, gh.

04 Alice funds you with 10 PAS if balance < 1.

05 Revive.map_account ties your SS58 to an H160.

06 Alice grants 1000 txs / 100 MB on Bulletin.

E2E Build and Deploy

dot deploy

```
Connecting to: wss://asset-hub-paseo-rpc.n.dwellir.com
SS58 Address: 5DfhGyQdFobKM8NsWvEeAKk5EQQgYe9AydGJ7rMB6E1EqRzV
H160 Address: 0x35cdb23ff7fc86e8dccc577ca309bfea9c978d20
```

```
Classifying name via PopOracle ...
Required status: ProofOfPersonhoodFull
Message: Requires Full personhood verification
```

```
dot deploy · kllljmm.dot · paseo v0.7.0
```

✓ deploy complete

```
url          https://kllljmm.dot.li
domain       kllljmm.dot
app cid      bafybeidwtjytbych3fxu4hwdvymslblzzcqsoivm6hpaywmccqidltn6ne
ipfs cid     bafybeigl5xdbepiduirgcnlxumane4a5prtcntcfd4clmj3m6tlcpjsvu
metadata cid bafk2bzacec7fthepbcq75cvno4l72d6mvvxkgs5xhkarewmo6mmkwkd7wfppe
```

`--signer dev` uses shared pool keys for DotNS registration. `--signer phone` signs every step with your key (needs 3–4 approvals).

`dot deploy` Runs in interactive mode and sets up everything and takes a confirmation before final deploy.

Run `dot deploy --help` to see all available options.

Signer and Ownership

step	signer	who owns it
bulletin upload	bulletin-deploy pool key	no owner • content-addressed
DotNS register	--signer dev or phone	whoever signed the tx
DotNS contenthash	same signer as register	same H160
registry publish	always you	env::caller() = your H160

Bulletin chunks go up under a shared pool key, nothing personal there. DotNS you choose: dev signer for speed, phone signer if you want the name under your key. The registry publish is non-negotiable — we force your signer so the contract records `env::caller()` as owner. That row is what `my apps` filters by.

Modify or remix other apps

```
dot mod · kkljjmm.dot · paseo v0.7.0
```

```
modding kkljjmm.dot
```

```
✓ fetch app metadata  
✓ fork & clone  
⚠ run setup.sh no setup.sh found
```

```
→ stuff
```

```
Next steps:
```

1. cd stuff
2. edit with claude
3. dot deploy --playground

`dot mod` looks up the app in the registry, finds the GitHub repo recorded at publish time, and forks it into your account. No `gh` login? It falls back to a fresh-history clone.

Then: change a color, change logic, change the whole frontend. Re-run `dot deploy`, you're the owner of a new entry in the feed.

pba-playground.dot

A React + PVM-contract app that reads every row of the Playground registry.

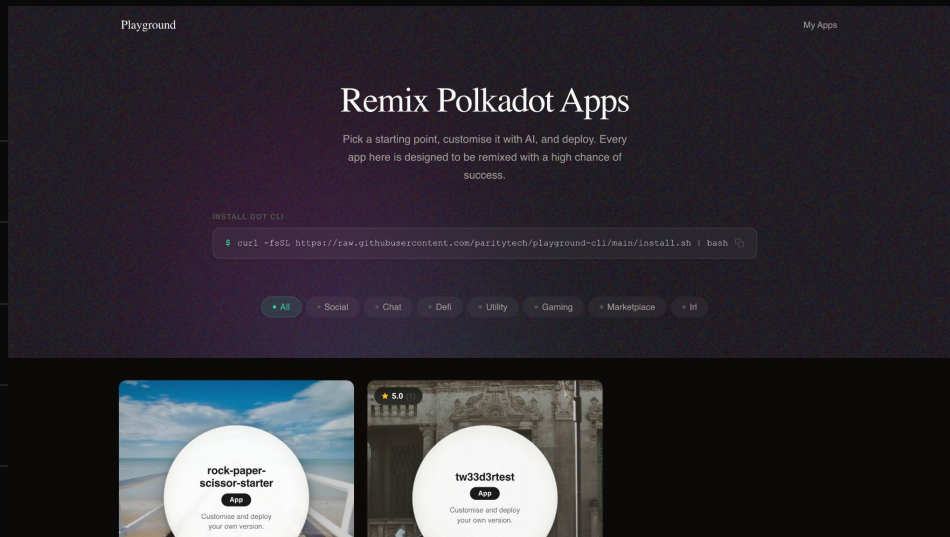
Infinite scroll, sorted by recency.

My apps filters by `env::caller()`.

Open → Polkadot Desktop / Mobile.

Remix → `dot mod <domain>`

Registry contract is open; anyone can read or publish.



Things you might see. And what to do.

WS halt (3)

Bulletin client timeout, rerun dot deploy

Out of PAS

rerun dot init to refund

Name reserved for governance

Pick a new domain

Out of memory on deploy

`DOT_MEMORY_TRACE=1 DOT_DEPLOY_VERBOSE=1`, file a bug

Deploy your contracts

Contracts are coming to dot deploy, meanwhile deploy them yourself

README size limits

README size limited to 20 Kb for rendering in the Playground app

Dev Signer overwrites

Use phone signer for DotNS interactions

If something else breaks: `DOT_DEPLOY_VERBOSE=1 DOT_MEMORY_TRACE=1 dot deploy 2> deploy.log` and create an issue.

Go build something truly Web3.

repos

github.com/paritytech/polkadot-apps

<https://github.com/paritytech/contract-dependency-manager>

<https://docs.polkadot.com/smart-contracts/overview/>

github.com/paritytech/playground-cli

github.com/paritytech/playground-app

ping

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